Introduction to Nutanix and AOS Training

Day -1

Strategy and Vision

When Nutanix was conceived it was focused on one goal: Make infrastructure computing invisible, anywhere

This simplicity was to be achieved by focus in three core areas:

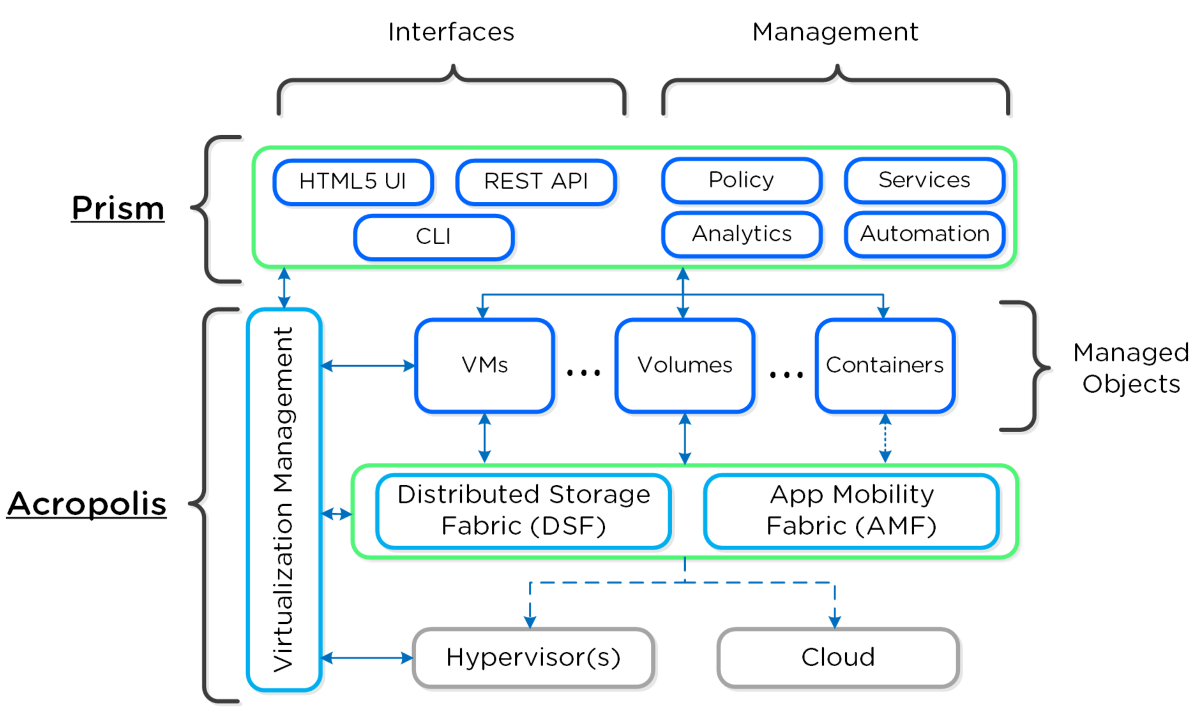
1. Enable choice and portability (HCI/Cloud/Hypervisor)

2. Simplify the “stack” through convergence, abstraction and intelligent software (AOS)

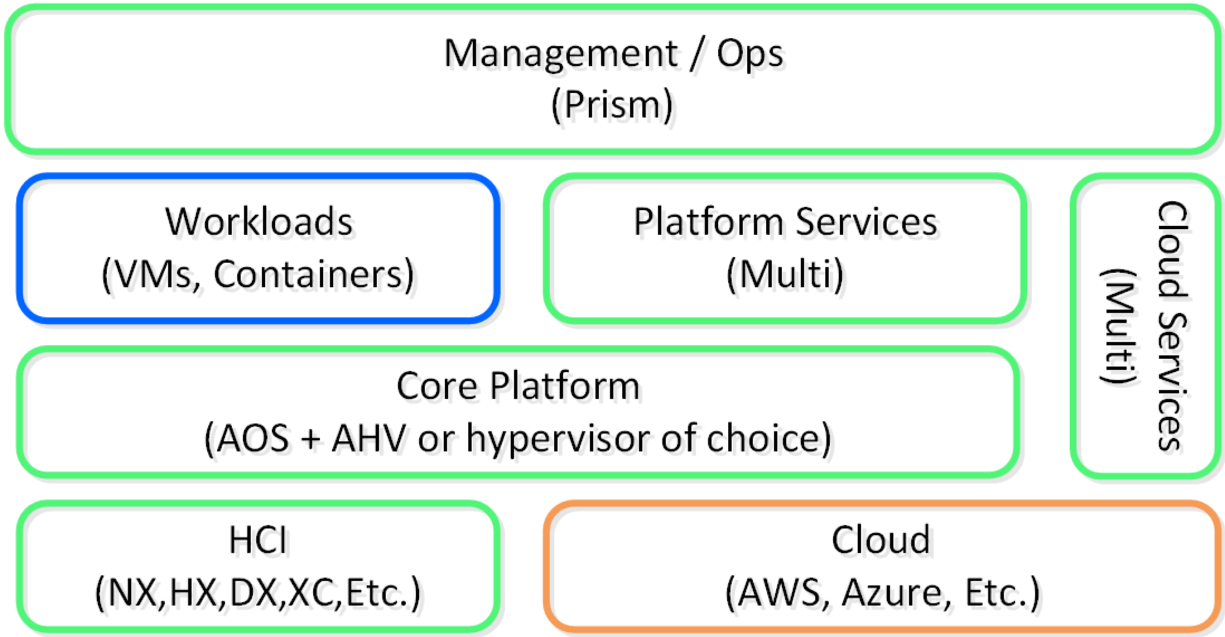
3. Provide an intuitive user interface (UI) through focus on user experience (UX) and design (Prism)

The following shows a high-level architecture of the Nutanix platform:

Constructs:

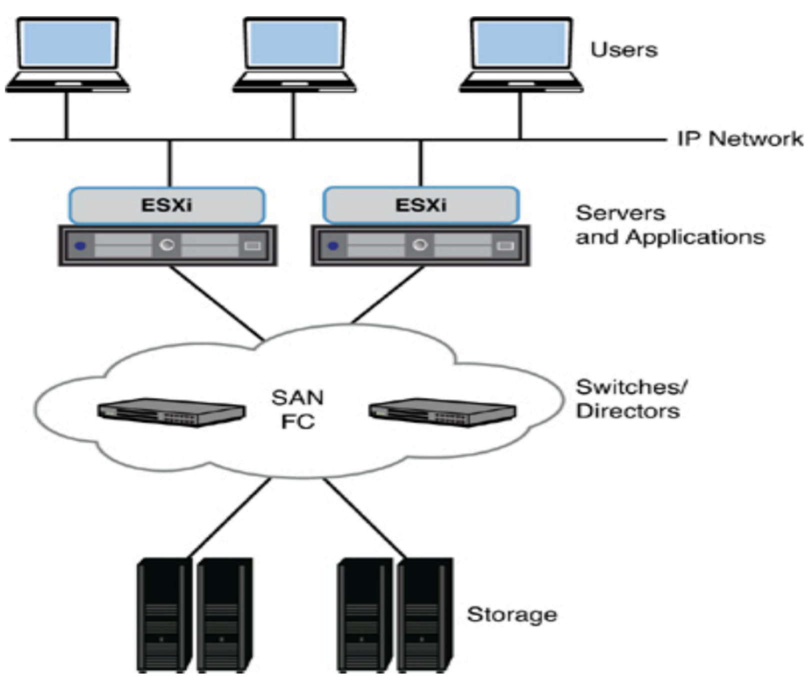


Architecture:



3-Tier Architecture

The 3-tier architecture follows a traditional model where the storage, compute, and network components are separate and independently managed. Each tier has its own dedicated hardware and management systems.



Challenges with 3-Tier Architecture

- High capital and managing cost.
- Inefficient Resource usage: Under utilization of compute usage
- Inherent Complexity: Features like vMotion, HA, and DRS led to the requirement of centralized storage
- High capital investment and IT infrastructure management cost
- Painful Management: Scalability, maintenance, software upgrade
- Invention of high speed storage lead to network bottleneck

HCI - Hyper Converger Infrastructure

Hyper-converged infrastructure (HCI) combines common datacenter hardware using locally attached storage resources with intelligent software to create flexible building blocks that replace legacy infrastructure consisting of separate servers, storage networks, and storage arrays.

Companies Providing HCI solutions: Nutanix, DELL VxRail, VMware vSAN, HPE SimpliVity, Cisco Hyperflex

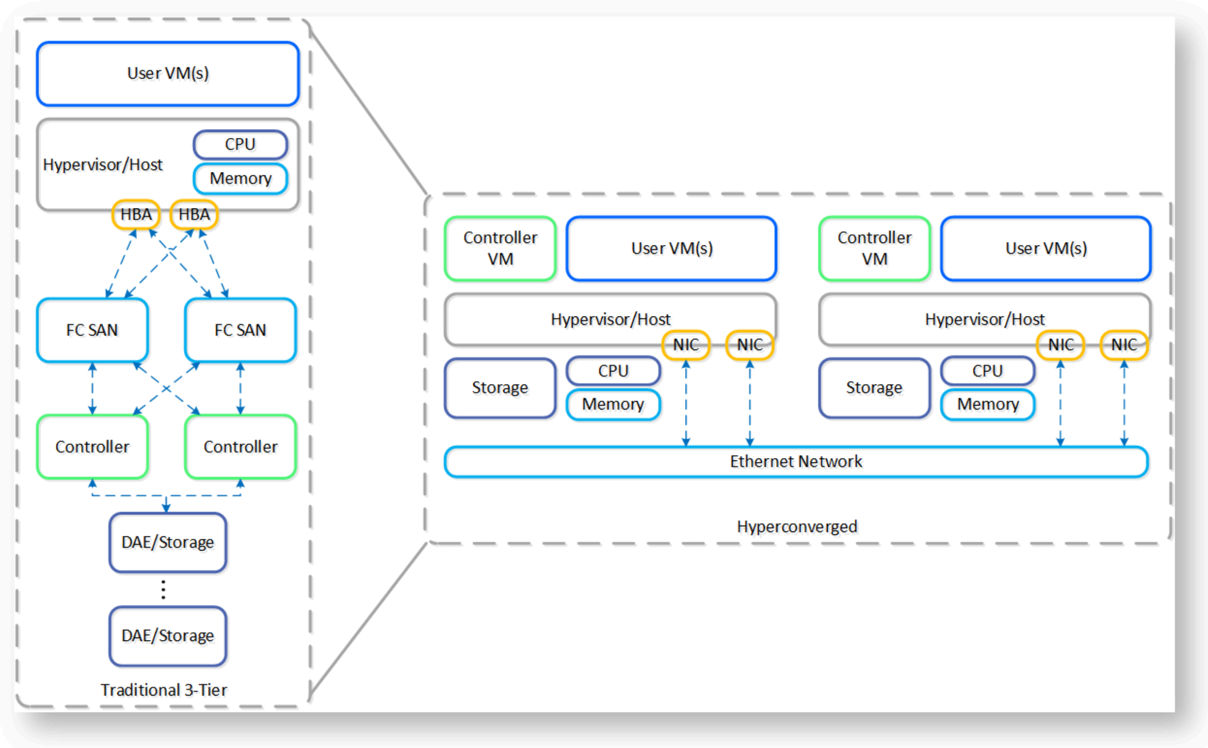
Core Constructs of Nutanix HCI Systems

- Must converge and collapse the computing stack (e.g. compute + storage)
- Must share (distribute) data and services across nodes in the system
- Must appear and provide the same capabilities as centralized storage (e.g. HA, live-migration, etc.)
- Must keep data as close to the execution (compute) as possible.
- Should be hypervisor agnostic
- Should be hardware agnostic

The Nutanix solution is a converged storage + compute solution which leverages local components and creates a distributed platform for running workloads.

How Nutanix HCI Works: https://youtu.be/6O7-nzXPAzs

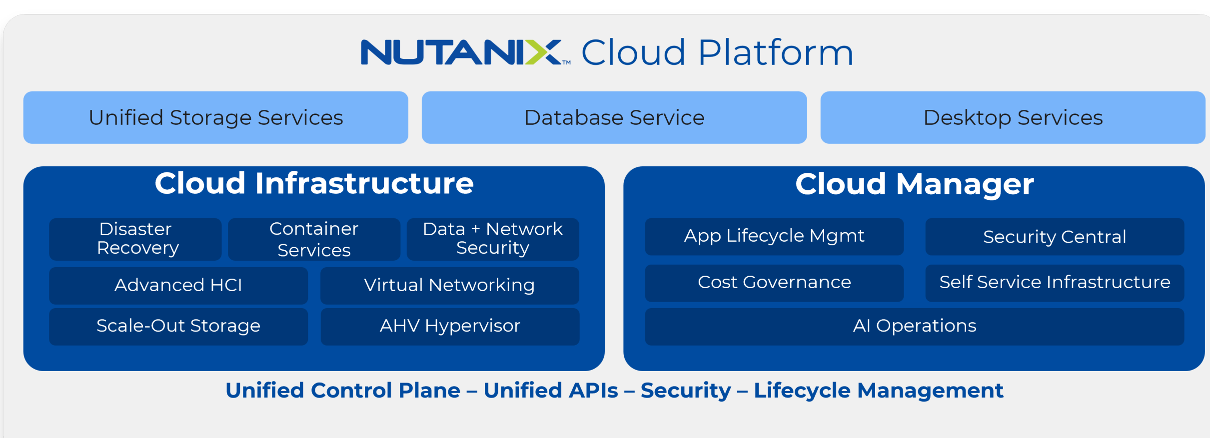
3-Tier vs HCI



As you can see, the hyperconverged system does the following:

- Virtualizes and moves the controllers to the host
- Provides core services and logic through software
- Distributes (shards) data across all nodes in the system
- Moves the storage local to the compute

Products and Platforms



Nutanix Cloud Platform

The Nutanix Cloud Platform is a secure, resilient, and self-healing platform for building your hybrid multicloud infrastructure to support all kinds of workloads and use cases across public and private clouds, multiple hypervisors and containers, with varied compute, storage, and network requirements.

The building blocks of Nutanix Cloud Platform are Nutanix Cloud Infrastructure and Nutanix Cloud Manager. These are the products that fall under each of them to form a complete solution:

Nutanix Cloud Infrastructure (NCI)	**Nutanix Cloud Manager (NCM)**
• AOS Scale-Out Storage ◦ This is the core of Nutanix Cloud Platform and provides a distributed, performant, and resilient storage platform that scales linearly. • AHV Hypervisor ◦ Native enterprise class virtualization, management and monitoring capabilities are provided by AHV. The Nutanix Cloud Platform also supports ESXi and Hyper-V. • Virtual Networking ◦ AHV comes with standard VLAN-backed virtual networking. You can also enable Flow Virtual Networking to provide virtual private cloud (VPC) and other advanced networking constructs in AHV for enhanced isolation, automation, and multi-tenancy. • Disaster Recovery ◦ Integrated disaster recovery that is simple to deploy and easy to manage, providing flexible RPO and RTO options on-prem and in the cloud • Container Services ◦ Nutanix Cloud platform provides compute and data for container services like OpenShift and provides an enterprise Kubernetes management solution (Nutanix Kubernetes Engine), to deliver and manage an end-to-end production ready Kubernetes environment. • Data and Network Security ◦ Comprehensive security for data using encryption with a built-in local key manager and software-based firewalls for network and applications with Flow Network Security.	• AI Operations (NCM Intelligent Operations) ◦ Optimizes capacity, proactively detects performance anomalies and provides capability to automate operational tasks that can streamline infrastructure management • Self-Service Infrastructure/App Lifecycle Management (NCM Self-Service) ◦ Orchestration with the ability to manage, deploy and scale applications across hybrid clouds with self-service, automation, and centralized role-based governance. • Cost Governance ◦ Drive financial accountability with intelligent resource sizing and accurate visibility into cloud metering and chargeback. • Security Central ◦ Security dashboard that unifies cloud security operations, identifies workload vulnerabilities, manages microsegmentation, and adheres to regulatory compliance requirements for strategic initiatives like Zero Trust.

Nutanix Unified Storage Services	**Nutanix Database Service - NDB**	**Desktop Services - Frame**

• Files Storage ◦ Simple, Secure, software-defined scale-out file storage and management • Objects Storage ◦ Simple, Secure and scale-out S3 compatible object storage at massive scale • Volumes Block Storage ◦ Enterprise class scale-out block storage that exposes storage resources directly to virtualized guest operating systems or physical hosts • Mine Integrated Backup ◦ Natively integrated turnkey backup solution	Simplified and automated database lifecycle management across hybrid clouds	Deliver virtual desktops and applications on any cloud at scale

Anatomy of a Node

Every node (NX, OEM, 3rd-party) is equipped with a variety of storage:

- SATADOM / M.2 card / NVME Boot drive - This is a module connected to the node's motherboard which presents a SATA or PCI device for storage. The hypervisor OS is installed here, along with a small ISO which allows the CVM to boot.
- SSD(s) - the CVM OS resides on SSD. If there are more than one SSDs, 3 software RAID partitions are created. The balance of SSD space is dedicated to oplog, metadata, and extent-store
- NVMe(s) - if the node is equipped with NVMe devices, these are treated as hot-tier and contain the node's Oplog
- HDD(s) - spinning disks in the node store a small amount of Curator data, with the rest being allocated to extent store

Node Architecture

Disks	Hybrid Disk Configuration, All-flash configuration, NVMe support
Storage Controller	LSI controller which utilizes PCI Passthrough on Hypervisor
CPU	G4 - Haswell G5 - Broadwell G6 - Skylake G7 - Cascade lake G8 - Icelake G9-Sapphire Rapids, G10 - Emerald rapids,AMD
Host Boot Drive	SATADOM, SATA M.2, NVMe
Network	Two 1gig Interfaces, and two 10gig interfaces,

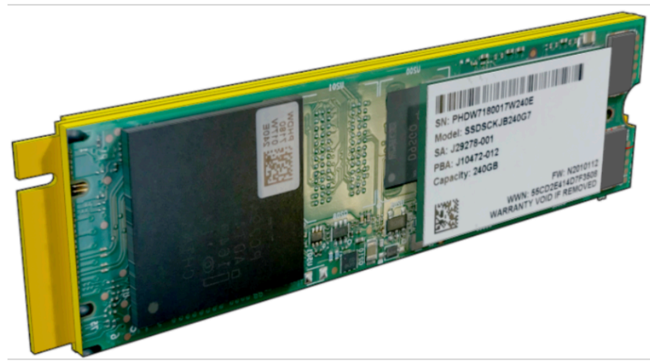
Hypervisor Boot Drives

G5 - SATADOM

G6 - M.2 SATA Drives

G7 - M.2 SATA Drives with RAID1 Mirroring

G8, G9, G10- NVMe Drives with RAID1 Mirroring



Lab exercise: Show AHV and ESXi hypervisors by logging into them.

AHV	**ESXi**	Hyperv
In AHV, the SATADOM is simply viewed as a disk, with the entire OS installed on one partition. In the latest versions, there are many other partitions that are created:	In ESXi, the SATADOM is always labelled as the ESXi local datastore, following the naming structure of "NTNX-local-ds-BlockSN-Position", or NTNX-local-ds-15SM65520205-A We can check the contents by navigating to the VMFS datastore for the local ds.	In Hyper-V, the SATADOM is partitioned into two disks, C: and D:
[root@Raka-AHV-BLR-2 ~]# df -h Filesystem Size Used Avail Use% Mounted on devtmpfs 252G 0 252G 0% /dev tmpfs 252G 34M 252G 1% /dev/shm tmpfs 252G 3.7M 252G 1% /run efivarfs 512K 187K 321K 37% /sys/firmware/efi/efivars /dev/mapper/ahv-root 4.8G 2.4G 2.2G 53% / /dev/mapper/ahv-tmp 966M 1.4M	[root@ESXi-Host-1:~] cd /vmfs/volumes/NTNX-local-ds-15SM65520205-A/ServiceVM_Centos/ [root@ESXi-Host-1:/vmfs/volumes/591b0b38-a273f7fc-abaa-0cc47a64f128/ServiceVM_Centos] ls -l total 209936 -rw------- 1 root root 0 Sep 30 08:38 ServiceVM_Centos-bbc9fa21.vswp -rw-rw-rw- 1 root root 586322 Oct 2 15:41 ServiceVM_Centos.0.out -rw-r----- 1 root root 28299264 Sep 30	nutanix@NTNX-15SM65520240-B-CVM:10.63.30.112:~$ winsh 192.168.5.1> df -h Filesystem Size Used Avail Use% Mounted on C:/cygwin/bin 59G 25G 34G 43% /usr/bin C:/cygwin/lib 59G 25G 34G 43% /usr/lib C:/cygwin 59G 25G 34G 43% / C: 59G 25G 34G 43% /cygdrive/c

899M 1% /tmp /dev/mapper/ahv-var 2.9G 141M 2.6G 6% /var /dev/nvme0n1p2 966M 128M 773M 15% /boot /dev/mapper/ahv-home 966M 108K 900M 1% /home /dev/mapper/ahv-varlog 457G 4.4G 429G 2% /var/log /dev/mapper/ahv-varlogaudit 966M 82M 819M 10% /var/log/audit /dev/nvme0n1p1 200M 5.8M 195M 3% /boot/efi tmpfs 51G 0 51G 0% /run/user/0 tmpfs 51G 0 51G 0% /run/user/101	08:36 ServiceVM_Centos.iso -rw-rw-rw- 1 root root 8684 Oct 9 09:31 ServiceVM_Centos.nvram -rw-rw-rw- 1 root root 0 May 16 14:27 ServiceVM_Centos.vmsd -rw-r----- 1 root root 4514 Sep 30 08:38 ServiceVM_Centos.vmx -rw------- 1 root root 0 Sep 30 08:38 ServiceVM_Centos.vmx.lck -rw-rw-rw- 1 root root 0 May 16 14:27 ServiceVM_Centos.vmxf -rw-r--r-- 1 root root 208173 May 16 14:31 vmware-1.log -rw-r--r-- 1 root root 214089 May 16 17:57 vmware-2.log -rw-r----- 1 root root 326872 Aug 15 17:35 vmware-3.log -rw-r----- 1 root root 210145 Sep 30 08:38 vmware-4.log -rw-r----- 1 root root 182975 Oct 6 09:21 vmware.log -rw------- 1 root root 178257920 Sep 30 08:38 vmx-ServiceVM_Centos-3150576161-1.vswp	D: 1020M 732M 289M 72% /cygdrive/d
The CVM data is stored in following locations: /var/lib/libvirt/NTNX-CVM contains the svmboot.iso file /etc/libvirt/qemu contains the xml definition for the CVM /var/log/libvirt/qemu contains the log file for the CVM /tmp/NTNX.serial.out.0 is the console log for the CVM	On a SATADOM, there will be a directory for CVM on it (which should *only* ever be a CVM) and this CVM directory will be called "ServiceVM_Centos". vmware.log = troubleshooting log specifically for this VM. SerivceVM_Centos.vmx = Configuration file containing the VM details for the CVM. ServiceVM_Centos.iso = Sometimes referred to as "svmboot" or "svmboot.iso", this boots the CVM into the proper partition on the SSD. ServiceVM_Centos.0.out = Console output for the CVM	

Nutanix Controller VM

The Nutanix CVM is responsible for the core Nutanix platform logic and handles services like:

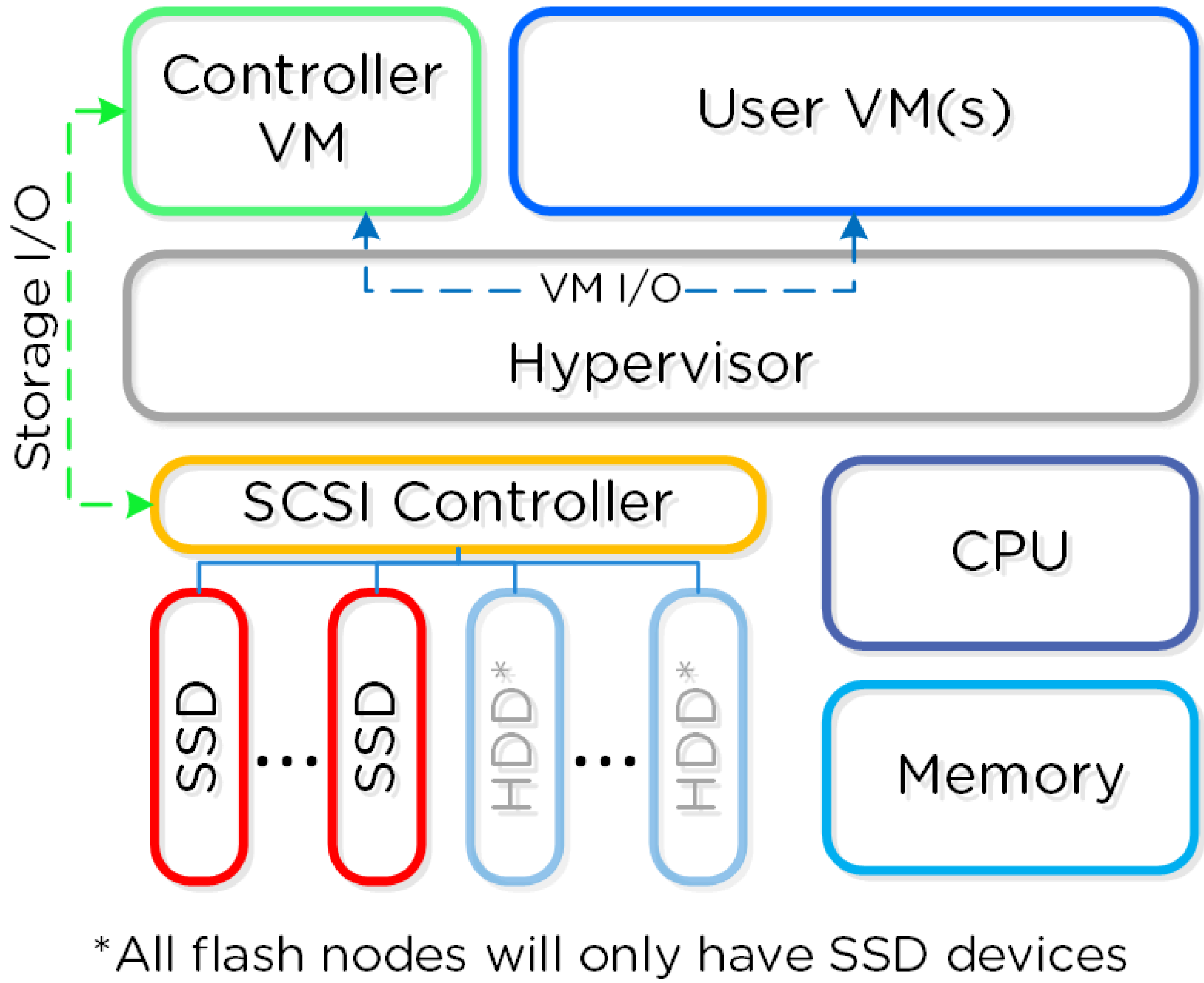
- Storage I/O & transforms (Deduplication, Compression, EC)

- UI / API
- Upgrades
- DR / Replication
- Etc.

The key reasons for running the Nutanix controllers as VMs in user-space really come down to four core areas:

1. Mobility
2. Resiliency
3. Maintenance / Upgrades
4. Performance, yes really

By running as a VM in user-space it decouples the Nutanix software from the underlying hypervisor and hardware platforms. This enabled us to rapidly add support for other hypervisors while keeping the core code base the same across all operating environments (on-premises & cloud). Additionally, it gave us flexibility to not be bound to vendor specific release cycles.



Anatomy of a CVM:

A CVM is a Linux VM with Nutanix services running inside it. It's responsible for core Nutanix Platform logic and handles services.

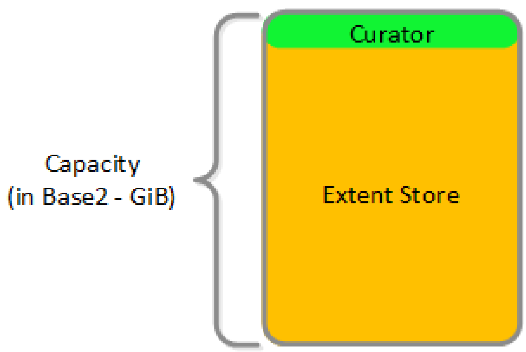
CVM's OS resides on SSD, along with space dedicated to Oplog, metadata and extent store

The CVM boots off of one of two boot partitions, depending on which one is marked as active. The active partition will be designated by having a file named `.G` located in the root directory.

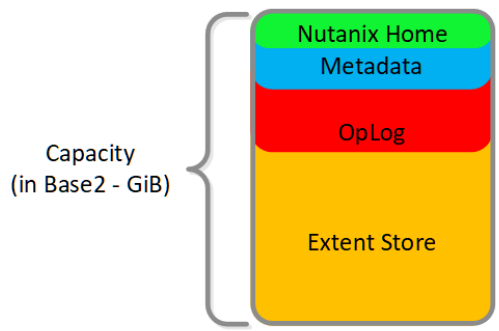
Mount Point	Size (AOS < 5.19)	Size (AOS 5.19 +)	Single-SSD Partition	Dual-SSD Partition	Colors in the diagram
/	10 GiB	12 GiB	/dev/sda1	/dev/md0	green
/ alternative	10 GiB	12 GiB	/dev/sda2	/dev/md1	green

/home	40 GiB	50 GiB	/dev/sda3	/dev/md2	green
Data & metadata	Rest of space	Rest of space	/dev/sda4	/dev/sd{a,b}4	light blue + red + dark blue + yellow

HDD



SSD



Single SSD:

```
nutanix@NTNX-15SM65520206-A-CVM:10.63.30.106:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       9.8G  6.9G  2.8G  72% /
devtmpfs         16G     0   16G   0% /dev
tmpfs           512M  4.0K  512M   1% /dev/shm
tmpfs            16G  888K   16G   1% /run
tmpfs            16G     0   16G   0% /sys/fs/cgroup
/dev/loop0      240M  2.4M  221M   2% /tmp
/dev/sda3        40G   25G   15G  63% /home
/dev/sdc1       5.5T  320G  5.1T   6% /home/nutanix/data/stargate-stora
ge/disks/NAHNRP6X
/dev/sdb1       5.5T  315G  5.1T   6% /home/nutanix/data/stargate-stora
ge/disks/NAHRWD0X
/dev/sda4       600G  330G  265G  56% /home/nutanix/data/stargate-stora
ge/disks/BTHC5456043Q800NGN
tmpfs           3.2G     0  3.2G   0% /run/user/1000
```

Dual SSD:

```
nutanix@NTNX-18SM6H020151-C-CVM:10.148.106.43:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
```

```
/dev/md0        9.8G  6.2G  3.5G  64% /
devtmpfs        9.8G     0  9.8G   0% /dev
tmpfs           512M  4.0K  512M   1% /dev/shm
tmpfs           9.9G  988K  9.9G   1% /run
tmpfs           9.9G     0  9.9G   0% /sys/fs/cgroup
/dev/loop0      240M  2.9M  221M   2% /tmp
/dev/md2         40G   21G   19G  52% /home
/dev/sde1       1.8T   86M  1.8T   1% /home/nutanix/data/stargate-stora
ge/disks/W460S43Y
/dev/sdc1       1.8T   87M  1.8T   1% /home/nutanix/data/stargate-stora
ge/disks/W460S3XJ
/dev/sdf1       1.8T   86M  1.8T   1% /home/nutanix/data/stargate-stora
ge/disks/W460PC8X
/dev/sdd1       1.8T   86M  1.8T   1% /home/nutanix/data/stargate-stora
ge/disks/W460S4CQ
/dev/sdb4       747G  212G  528G  29% /home/nutanix/data/stargate-stora
ge/disks/S3F3NX0J909995
/dev/sda4       747G  196G  544G  27% /home/nutanix/data/stargate-stora
ge/disks/S3F3NX0J909898
tmpfs           2.0G     0  2.0G   0% /run/user/1000


nutanix@NTNX-18SM6H020151-C-CVM:10.148.106.43:~$ cat /proc/mdstat
Personalities : [raid1]
md2 : active raid1 sdb3[1] sda3[0]
      41909248 blocks super 1.1 [2/2] [UU]
      bitmap: 1/1 pages [4KB], 65536KB chunk

md1 : active raid1 sda2[0] sdb2[1]
      10476544 blocks super 1.1 [2/2] [UU]
      bitmap: 0/1 pages [0KB], 65536KB chunk

md0 : active raid1 sda1[0] sdb1[1]
      10476544 blocks super 1.1 [2/2] [UU]
      bitmap: 1/1 pages [4KB], 65536KB chunk
```

HDD Usage

Since HDD devices are primarily used for bulk storage, their breakdown is much simpler:

- Curator Reservation (Curator storage)

- Extent Store (persistent storage)

In case of All Flash Systems

The first 2 SSDs in an all flash system will be partitioned the same way as the SSDs in a dual SSD system. The rest of the SSDs will be used as data disks and will have a single data partition. The additional SSDs will be designated as belonging to the SSD tier and contain OPLOG, metadata (Cassandra + AES) and Extent Store

CVM /home/nutanix Layout

Within the `/home/nutanix` directory on the CVM, there are several directories it is worthwhile to be familiar with:

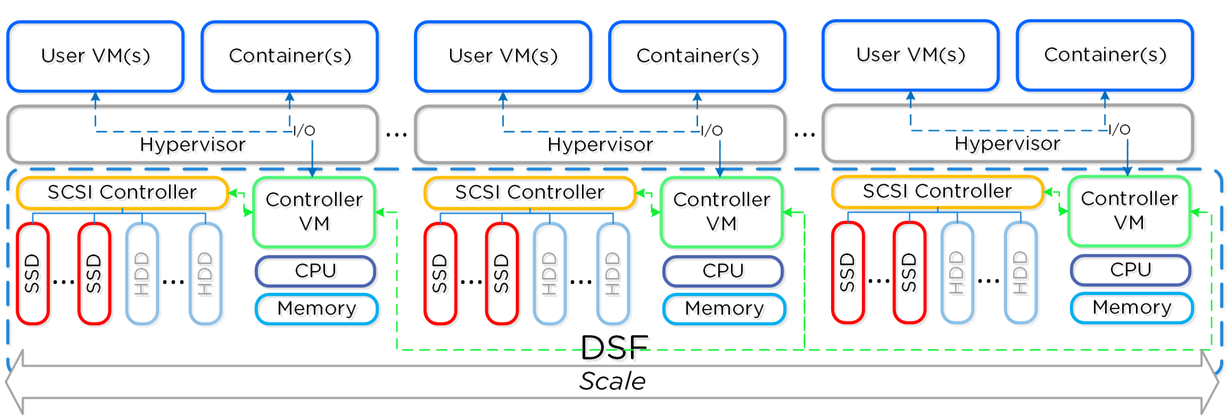
Directory	Purpose
~/data/logs	All Nutanix service logs
~/data/logs/sysstats	System and process statistics
~/data/stargate-storage/disks	Mount points for physical disks/partitions (Extent Store)
~/serviceability/bin	Scripts utilized automatically by the CVM or manually by SREs
~/config	Cluster configuration details and custom GFLAG settings

Distributed System

There are three core principles for distributed systems:

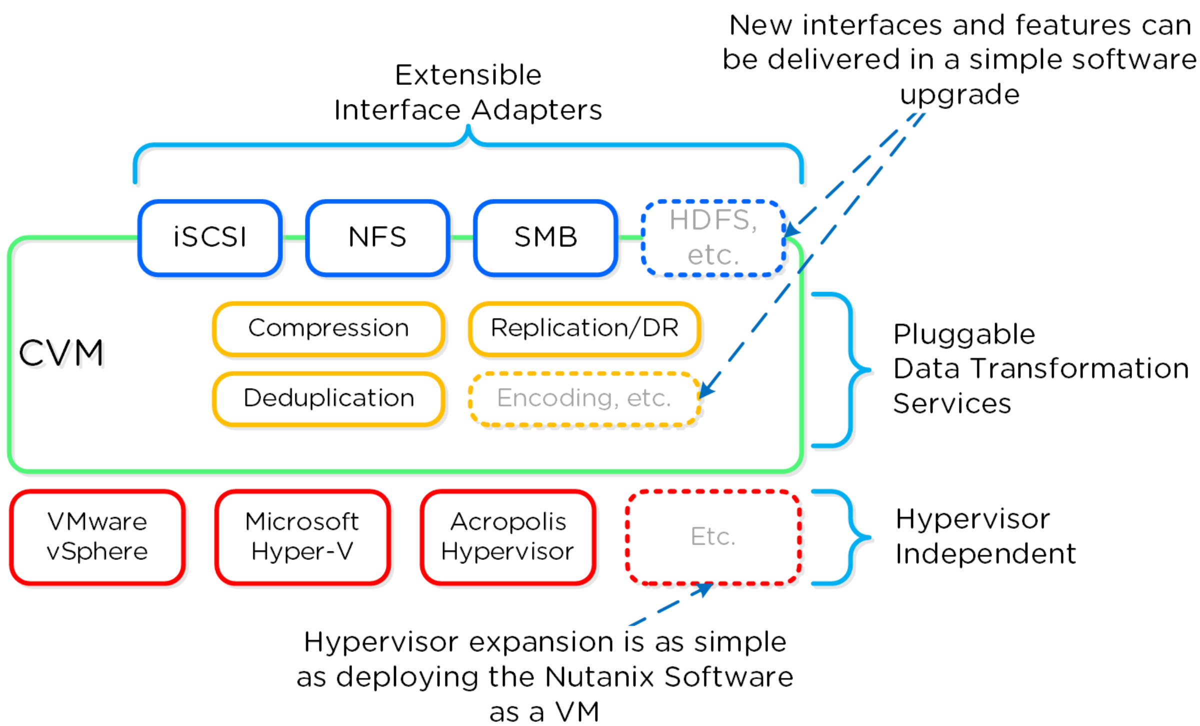
1. Must have no single points of failure (SPOF)
2. Must not have any bottlenecks at any scale (must be linearly scalable)
3. Must leverage concurrency (MapReduce)

Together, a group of Nutanix nodes forms a distributed system (Nutanix cluster) responsible for providing the Prism and AOS capabilities. All services and components are distributed across all CVMs in a cluster to provide for high-availability and linear performance at scale.



Software Defined

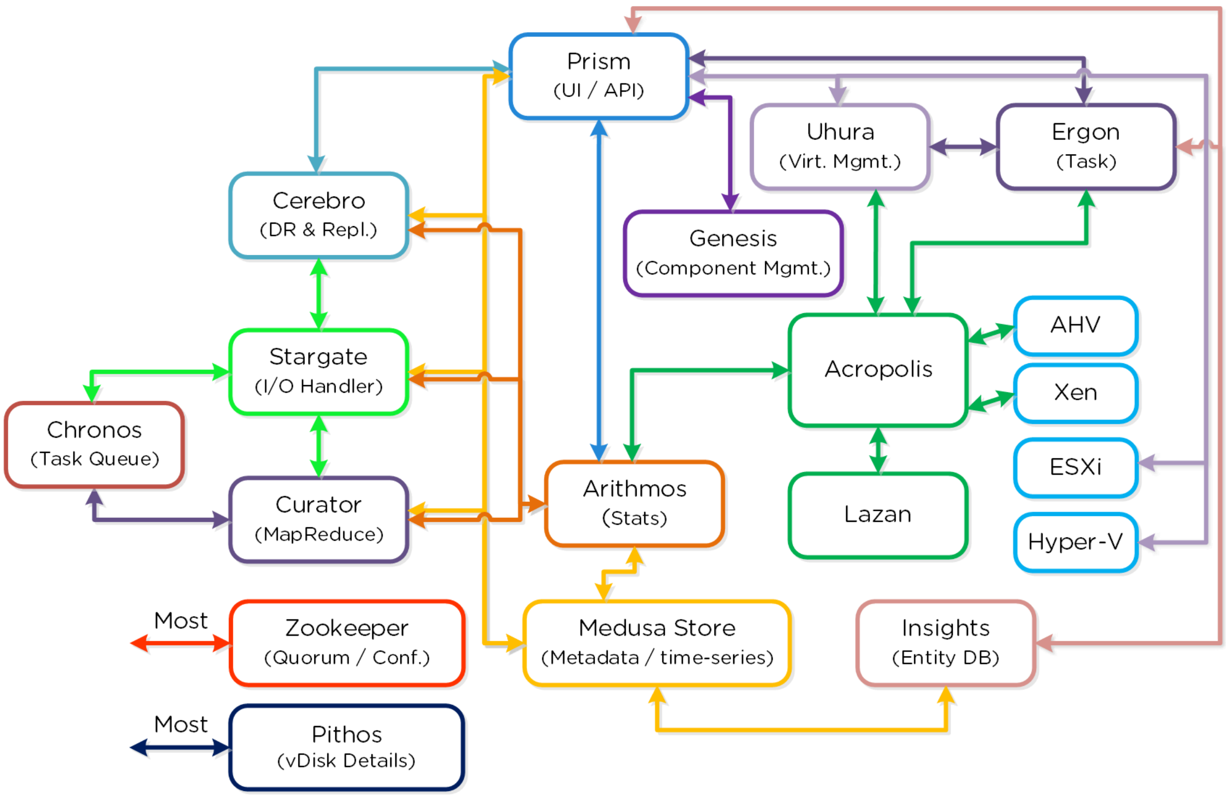
There are four core principles for software definition systems:



- Must provide platform mobility (hardware, hypervisor)
- Must not be reliant on any custom hardware
- Must enable rapid speed of development (features, bug fixes, security patches)
- Must take advantage of Moore’s Law

Since all features are deployed in software, they can run on any hardware platform, any hypervisor, and be deployed through simple software upgrades.

Cluster Components



Cassandra: Distributed metadata store

Zookeeper: Cluster configuration manager

Stargate: Data I/O manager

Curator: MapReduce cluster management and cleanup

Prism: UI and API

Genesis: Cluster component & service manager

Chronos: Job and task scheduler

Cerebro: Replication/DR manager

Pithos: vDisk configuration manager

Arithmos: Maintain historical stats and config information

Lazan: DRS, advanced scheduling in AHV clusters in AOS 5.0 and newer.

Uhura: VM management back-end service

Ergon: unified task management framework for all AOS services

PCI Passthrough

Disks are presented to the CVM using PCI passthrough. This allows the full PCI controller (and attached devices) to be passed through directly to the CVM and bypass the hypervisor.

If we check CVM's XML file on an AHV host, and look for the LSI controller that is passed to the CVM:

```
[root@Veerappan-1 ~]# virsh dumpxml 20 | grep -A8 \<hostdev
<hostdev mode='subsystem' type='pci' managed='yes'>
<driver name='vfio'/>
<source>
<address domain='0x0000' bus='0x01' slot='0x00' function='0x0'/>
</source>
<alias name='hostdev0'/>
<rom bar='off'/>
<address type='pci' domain='0x0000' bus='0x00' slot='0x06' function='0x
0'/>
</hostdev>
```

Checking the source bus address:

```
[root@Veerappan-1 ~]# lspci | grep LSI
01:00.0 Serial Attached SCSI controller: Broadcom / LSI SAS3008 PCI-Exp
ress Fusion-MPT SAS-3 (rev 02)
```

On CVM, the destination:

```
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ lspci | grep LSI
00:06.0 Serial Attached SCSI controller: Broadcom / LSI SAS3008 PCI-Exp
ress Fusion-MPT SAS-3 (rev 02)
```

CVM Boot Process

Why do we need svmboot.iso?

1. CVM boots from svmboot.iso

   ```
   [root@Tyagi-2 ~]# grep -i iso /root/NTNX-CVM.xml
   <source file="/var/lib/libvirt/NTNX-CVM/svmboot.iso"/>
   ```

2. Grub in svmboot.iso loads Kernel and initrd from the iso.

```
[root@Tyagi-2 ~]# mount /var/lib/libvirt/NTNX-CVM/svmboot.iso /mnt
mount: /dev/loop0 is write-protected, mounting read-only
[root@Tyagi-2 ~]#
[root@Tyagi-2 ~]# ls -al /mnt/boot/
total 5519
dr-xr-xr-x. 3 root root    2048 Dec 19 06:58 .
dr-xr-xr-x. 3 root root    2048 Dec 19 06:58 ..
r--r--r--. 2 root root  0 Dec 19 06:58 .done
dr-xr-xr-x. 3 root root    2048 Dec 19 06:58 grub
r--r--r--. 2 root root  427 Dec 19 06:58 grub.cfg
r--r--r--. 2 root root 3059748 Dec 19 06:58 initrd
r--r--r--. 2 root root 2584528 Aug 28 22:22 kernel
```

3. Grub asks the kernel to load all the required modules.
4. Grub asks the kernel to execute script(s) to find the active root partition in the SSD.
5. The script mounts the active root partition from the SSD.
6. Calls Kexec to switch from the kernel in the ISO to kernel in the SSD.
7. Kernel now loads initrd from the SSD and contents in SSD takes over.

More around this will be covered on the next day.

Memory allotted to a CVM

https://portal.nutanix.com/page/documents/details?targetId=Field-Installation-Guide-v5_2:fie-cvm-cpu-allocation.html

---

CVM Networking

Regardless of hypervisor, all nodes running AOS have few things in common:

- Internal vSwitch - for CVM and it's local Host communication
- External vSwitch - External vSwitches are used for communication between the CVMs and other CVMs, hosts, and any external networks

Internal vSwitch

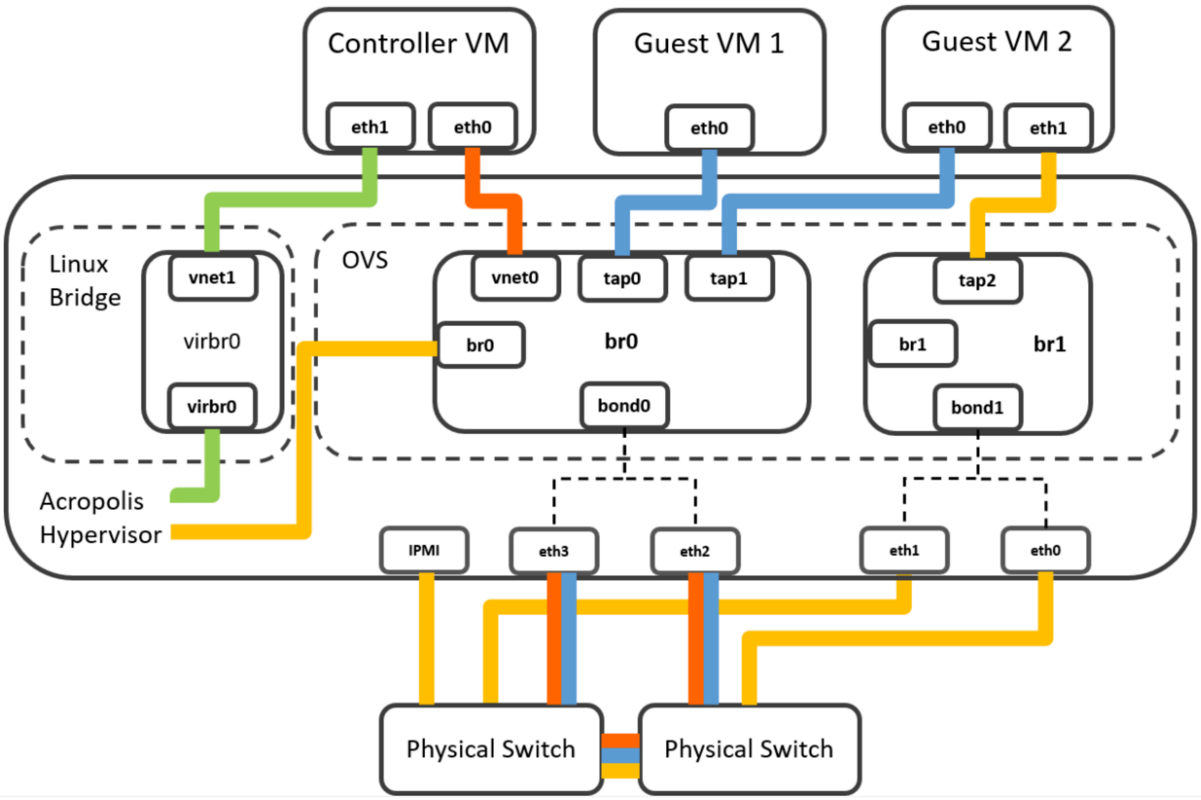
Every node has an "internal vSwitch" which is used exclusively for communication between the hypervisor and its local CVM. This internal switch utilizes the `192.168.5.x` network, detailed below:

Hypervisor	Netmask	Description
192.168.5.1	255.255.255.0	A dedicated IP address for the local Hypervisor on the internal vswitch assigned to the virbr0 interface on the Hypervisor. Use this IP address to SSH from the CVM to the host.
CVM		
192.168.5.2	255.255.255.128	Storage communication. This is assigned to CVM interface eth1. If the stargate process on the local CVM is down, traffic to this IP address will be redirected to another CVM in the cluster. This mechanism is described in the next section.
192.168.5.254	255.255.255.0	A dedicated IP address for the local CVM on the internal vswitch assigned to the eth1:1 sub-interface on the CVM. Traffic destined for this IP address is never redirected. Use this IP address to SSH from a host in the cluster to its local CVM.

- The Internal vSwitch should not be modified under any circumstance - it is defined when the node is imaged
- There are NO physical NICs attached to the internal vSwitch
- The `192.168.5.2` interface is used by the hypervisor to communicate with its local CVM for storage traffic (NFS, iSCSI, SMB) and for collecting metric data from the hypervisor
- The `192.168.5.254` interface should always be used to access the local CVM from the host/hypervisor. This is useful if the cluster is currently in a HA state.

Internal vSwitch in AHV

For AHV this internal vSwitch is a Linux Bridge titled `virbr0`



bondX vs. brX-up

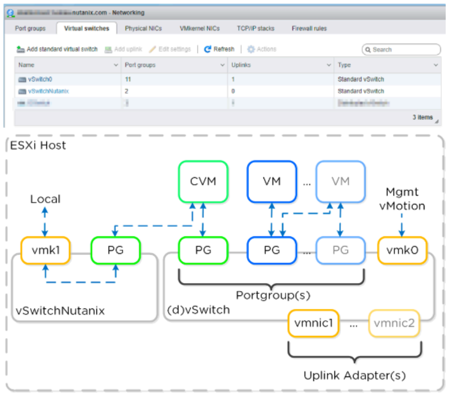
Depending on the AOS version instead of bond0 and bond1 one might see br0-up and br1-up.

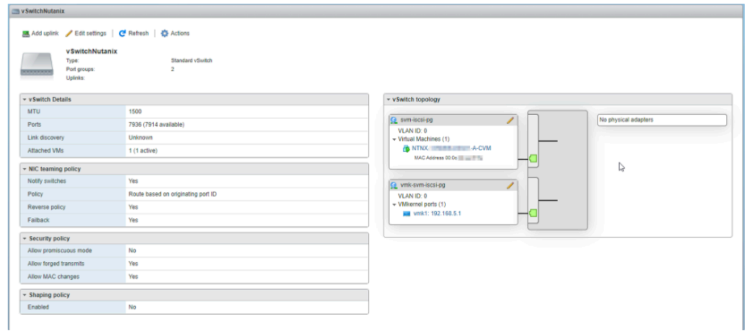
The naming convention can further change between AOS versions, but the architecture remains the same (bondX=brX-up and contain the uplinks for that OVS to the external world and physical network)

Internal vSwitch in ESXi

For ESXi this internal vSwitch is a standard switch called `vSwitchNutanix`. By default the below 2 switches are the only ones present on an ESX deployment of a Nutanix Cluster.

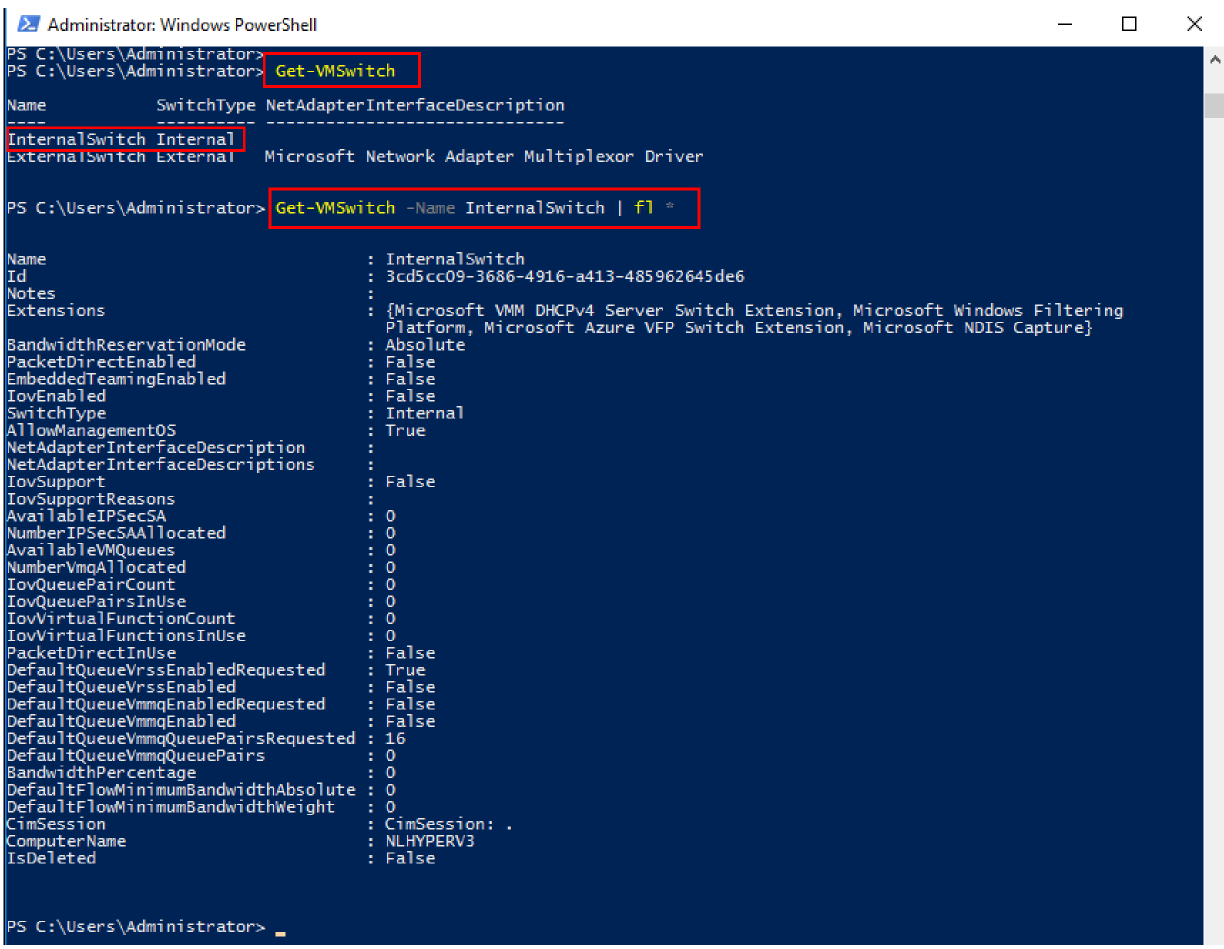
The host has a vmkernel interface on this vSwitch (vmk1 - 192.168.5.1) and the CVM has an interface bound to a port group on this internal switch (svm-iscsi-pg - 192.168.5.2). This is the primary storage communication path.





Internal vSwitch in Hyper-V

For Hyper-V this is an internal vSwitch is called `InternalSwitch`



External vSwitch

External vSwitches are used for communication between the CVMs and other CVMs, hosts, and any external networks. Customers are able to modify this switch to fit their needs. The details for how the external vSwitch is configured will be covered under each hypervisor.

Network Access

- The hosts and CVMs should always have SSH enabled. Services on a CVM will not start if SSH is disabled or authentication between the CVM and its local hypervisor fails.
- The foundation process configures pre-shared keys on each CVM for all other CVMs and all hypervisors/hosts in the cluster. Keys are not configured to allow password-less access from the hypervisor to the CVMs. Password-less access works only from between the CVMs themselves and between the CVMs and all the hosts
- IPMI interfaces have a completely separate network stack from the host, and can be configured via `ipmitool` on the corresponding host or in the BIOS during bootup.